

THOMAS BERNARD



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TECHNICAL SKILLS

- Fluvial morphodynamics | landslides
- Natural-hazards analysis | topographic change detection
- Hydraulic modelling 1D/2D | statistical modelling
- DTM/DSM creation
- 3D data processing (LiDAR topobathymetric, photogrammetry)

SOFTWARES & TOOLS

- Python (advanced) | Github | Machine & Deep learning
- HEC-RAS | Graphflood
- QGIS | CloudCompare | Lastools
- Microsoft suite

SOFT SKILLS

- Project & budget management
- International communication
- Autonomy, adaptability, fast-learner, responsible

LANGUAGES

French | Madrelingua English | C2
Italian | C1 German | A2

Researcher in Earth Sciences

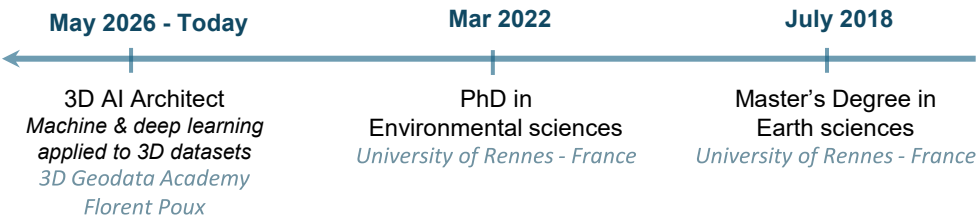
Expert in Natural-hazards, Hydromorphology and Geospatial analyses

I am an Environmental researcher and computational specialist with a PhD in Environmental Sciences and a strong foundation in natural-hazards and geospatial analysis. I am using advanced Python, modelling and complex 3D data processing to understand natural processes and assess related risks. Backed by extensive experience in 1D/2D hydraulic modelling, and recently enhanced by specialized training in machine and deep learning applied to 3D datasets, I am eager to apply my technical expertise to advance innovative climate risk solutions at Eurac Research.

PROFESSIONAL EXPERIENCES

- Jun 2026 - Jul 2026 Hydraulic engineer (consultant)**
Vesta Energy srl – Trento, Italy
- Simulation of 100-year floods to **assess the impact of infrastructure in flood-prone areas**
 - Technical standards of the PGRA 2021-2027
 - Responsible of the Hydraulic Compatibility Assessment Study
- Oct 2023 - Apr 2026 Post-doc researcher in Hydromorphology**
University of Trento - Italy
- Management of an applied research project in river morphodynamics
 - **1D/2D hydraulic and sediment transport modelling** (numerical and physical)
 - Preparation of technical reports and coordination with international partners
 - Organization of a technical workshop on the management of Alpine rivers (planning, funding, coordination)
- Apr 2022 - Jan 2023 Remote sensing & Geospatial engineer**
University of Rennes - France
- Processing and classification of **topo-bathymetric LiDAR** data using **machine learning**
 - Reconstruction of river bathymetry using **2D hydraulic simulations**
 - Production of high-resolution digital terrain models (DTMs)
- Oct 2018 - Mar 2022 Junior researcher in Natural-hazards and Remote sensing**
University of Rennes - France
- 2D hydraulic simulations for **floodplain mapping and risk analysis**
 - Development of workflows for detecting **topographic changes** and landslides
 - LiDAR data processing and classification
 - Production of technical reports | international communication
 - Teaching activity >100 hours

EDUCATION



SCIENTIFIC PUBLICATIONS

- **2026 (in review):** Nativ, R., Lague, D., Leroy, P., Guerit, L., **Bernard, T.**, Letard, M., ... & Steer, P. (2026). From point cloud to digital elevation model: Airborne topo-bathymetric LiDAR processing over the 28 km Ardèche River Gorges. *EGUsphere*, 2026, 1-47.
- **2026 :** **Bernard, T. G.**, Pandrin, E., Capito, L., Bizzi, S., & Bertoldi, W.: "*Quantification of the active width in gravel-bed rivers: the effect of morphology, flood magnitude and survey frequency*". *Geomorphology*.
- **2024 :** Gailleton, B., Steer, P., Davy, P., Schwanghart, W., & **Bernard, T.** (2024). GraphFlood 1.0: an efficient algorithm to approximate 2D hydrodynamics for landscape evolution models. *Earth Surface Dynamics*, 12(6), 1295-1313.
- **2022 :** **Bernard, T. G.**, Davy, P., & Lague, D. : "*Hydro-geomorphic metrics for high resolution fluvial landscape analysis*". *Journal of Geophysical Research: Earth Surface*, 127(3), e2021JF006535. (**Top cited article** of 2022-2023)
- **2021 :** **Bernard, T. G.**, Lague, D., & Steer, P. : "*Beyond 2D landslide inventories and their rollover: synoptic 3D inventories and volume from repeat lidar data*". *Earth Surface Dynamics*, 9(4), 1013-1044.

PhD thesis: Bernard, T. (2022). "*Analyse haute résolution de la morphologie des paysages et des processus à partir de LiDAR aéroporté répété et simulation hydraulique*" (Doctoral dissertation, University of Rennes).

INTERNATIONAL CONFERENCES

- **July 2025 (Lyon, France) :** Bernard, T., Pandrin, E., Bertoldi, W., Capito, L. & Bizzi, S.: "*Bedload and morphological active widths of gravel-bed rivers: variability among timescales*". In *EGU General Assembly Conference Abstracts* (pp. EGU25-4044).
- **April 2025 (Vienna, Austria) :** Bernard, T., Pandrin, E., Bertoldi, W., Capito, L. & Bizzi, S.: "*Variability in bedload and morphological active widths of gravel-bed rivers across timescales*". In *EGU General Assembly Conference Abstracts* (pp. EGU25-4044).
- **April 2021 (Vienna, Austria):** Bernard, T., Davy, P., & Lague, D. : "*A hydrodynamically consistent" slope-area" relationship for analysing fluvial landscape with wide rivers*". In *EGU General Assembly Conference Abstracts* (pp. EGU21-8423).
- **December 2019 (San Francisco, United-States):** Bernard, T., Davy, P., & Lague, D. : "*How 2D Hydraulic Modelling Can Improve Landscape Analysis in Tectonic Geomorphology?*". In *AGU Fall Meeting Abstracts* (Vol. 2019, pp. EP31C-2301).
- **April 2019 (Vienna, Austria):** Bernard, T., Lague, D., & Steer, P. (2019, January). "*Detection and volume estimation of earthquake-triggered landslides from multitemporal LiDAR and tri-stereo satellite 3D remote sensing*". In *Geophysical Research Abstracts* (Vol. 21).

